Fiscal Year:	FY 2016	**Task Last Updated:**	FY 06/14/2016
PI Name:	Venkateswaran, Kasthuri Ph.D.		
Project Title:	ISS Microbial Observatory - a Genetic Approach		
Division Name:	Space Biology		
Program/Discipline:	SPACE BIOLOGY		
Program/Discipline--Element/Subdiscipline:	SPACE BIOLOGY--Cellular and molecular biology		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	(1) Cell & Molecular Biology (2) Microbiology		
Space Biology Cross-Element Discipline:	(1) Reproductive Biology		
Space Biology Special Category:	None		
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Zip Code:	91109-8001	**Congressional District:**	27
Comments:			
Project Type:	Flight	**Solicitation / Funding Source:**	2012 Space Biology NNH12ZTT001N
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No. of Post Docs:	3	**No. of PhD Degrees:**	
No. of PhD Candidates:		**No. of Master' Degrees:**	
No. of Master's Candidates:		**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:	4	**Monitoring Center:**	NASA ARC
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Flight Program:	ISS		
Flight Assignment:	ISS Microbial Observatory NOTE: Extended to 12/31/2017 per F. Hernandez/ARC (Ed., 8/22/17) NOTE: Extended to 04/30/2017 per F. Hernandez/ARC (Ed., 2/2/17) NOTE: Extended to 12/31/2016 per F. Hernandez/ARC (Ed., 4/11/16) NOTE: Extended to 4/30/2016 per A. Chu/ARC (Ed., 9/17/15)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Fox, George (University of Houston) Pierson, Duane (NASA Johnson Space Center)		
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Task Description:	As recommended by the National Research Council Decadal Survey, the Jet Propulsion Laboratory (JPL) proposes to establish an International Space Station-Microbial Observatory (ISS-MO) to generate a microbial census of the space station"s surfaces and atmosphere using advanced molecular microbial community analysis techniques, supported by traditional culture-based methods and modern bioinformatic computational modeling. The proposed ISS-MO establishment will lead to long-term, multigenerational studies of microbial population dynamics, addressing recommendation P1 of this solicitation. It will also provide answers to NASA Space Biology Guiding Questions CMM-1, CMM-4, and CMM-5. The ISS-MO project's methodology will serve as the foundation for an extensive microbial census, offering significant insight into spaceflight-induced changes in the populations of beneficial and potentially harmful microbes. Additionally, it will provide NASA with both a mechanistic understanding of these changes (e.g., cataloging population changes and mapping/linking these to environmental niche and genomic changes), as well as insight into practical countermeasures for mitigating risks to humans and environmental systems. The ISS-MO team will use existing ISS sample collection technologies to generate an initial microbial census. Following their return to Earth, samples from the various ISS modules will be analyzed using standardized technologies from the Mars Program-funded projects. The proposed ISS-MO project includes the delivery of a database that compiles the genomic sequences and genetic information for all the microbes encountered within the ISS habitat. Using this data, NASA can more accurately and confidently assess the status of microbes associated with closed habitation and crew health maintenance. In addition to providing microbial profiles, the ISS-MO team will identify which microbial taxa pose particular threats to crew health. Furthermore, specific ISS-MO project aims will enable us to resolve applicable NASA-Human Research Program integrated research plan risks.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	The NASA crewmembers are exposed constantly to the dust in a closed system, and have also reported allergies. Subsequent microbiological analyses using traditional methods noticed no anomaly and were also not able to predict the causative microbial agents. The next-generation sequencing study provided NASA with the ability to assess the spectrum of microorganisms associated with the ISS surfaces and would help to predict possible microbial agents that might affect crew health. The results of the present study also exhibited presence of representative human associated (skin, gastrointestinal track, wounds, etc.) bacterial commensals and opportunistic fungal pathogens. Furthermore, based on the phylogenetic affiliation it is possible to determine whether these viable microorganisms are hardy and robust that have been previously reported to survive in oligotrophic environments (cleanrooms, hospital, commercial airliner, etc.). The study has improved our understanding of background microbial contamination, thus facilitating the development of biosensors to monitor closed habitats like ISS and future manned missions.
	• Samples were collected from two different flights at the same ISS locations. Each sampling event collected 8 surface and 6 air samples. • Cultivable bacteria/fungi were carried out and their identification using molecular phylogeny. • Antibiotic resistant pattern in cultivable Staphylococcus strains was carried out. • Total and viable microbial burden analysis were carried out using molecular methods. – ATP – assay for total and viable microbial burden ; – PMA – assay for total/viable bacterial burden and diversity • iTag Illumina – based sequencing was used for bacterial, archaeal, and fungal diversity. • Metagenome sequencing for all kinds of microorganisms is under way. Cultivable microbial burden 1. The cultivable microbial bioburden differed by location and sampling event. R2A is a low nutrient media that supports growth of environmental slow-growing bacteria. The location no. 6 (PMM, Port 1) did not yield any cultivable bacterial isolates in both samplings, while locations no. 3 (Node 3; ARED Foot platform) and no. 8 (Node 2; Port crew quarters Bump-Out exterior aft wall) did not yield any isolates in second and first sampling respectively. In other locations the bioburden ranged from 107-1010 CFU/m2 per sampling location. 2. Fungal population from PDA varied from 105-108 CFU/m2. Although mycoidal fungi and yeast isolates were retrieved from each location, they were not as abundant as environmental bacteria on the average. 3. Despite different yields for the locations during two consecutive samplings, location no. 3 and location no. 6 yielded less cultivable population than other surfaces. Viable bacterial and fungal burden as seen in molecular assays 4. The yields of bacterial DNA varied between the locations both for total and viable populations: 106-1010 and 105-1010 rrn copies per m2, respectively. 5. The samplings showed different yields for the same locations, although the trend was very similar: location no. 3 and 6 that resulted in lower viable populations in both samplings than other locations. 6. Cultivable population were less than 10% in most of the cases (8 out of 16 ISS surface measurements) and ~3 out of 16 ISS surface measurements showed that almost all cells could be cultured. 7. Viable bacterial population was measured via PMA-treated qPCR-based analyses and showed that >90% could not be cultures in 8 of 16 surface samples. About 5 out of 16 locations showed less than 1% bacteria were cultivable. Cultivable Bacterial Community Composition 8. The cultivable bacterial diversity belonged only to three phyla: Actinobacteria, Firmicutes, and Proteobacteria. 9. On the genus level, the most represented was Staphylococcus (31% of total bacterial isolates), Bacillus (14% of total

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bacterial isolates), Pantoea (11% of total bacterial isolates).

10. The most occurring species was Staphylococcus aureus and Pantoea conspicua (both constituted 11.8% of total bacterial isolates).

11. The high occurrence of S. aureus was correlated with the repeating sampling of the isolates from location no. 6 in both sessions. Similarly, P. conspicua was isolated only from the location no. 5 (Node 1 Overhead 4) in both samplings constituting over 90% cultivable isolates retrieved from this location.

Cultivable Fungal Community Composition

12. The fungal diversity was mostly represented by yeast isolates from species Rhodotorula mucilaginosa (51 % of total fungal isolates) and followed by Penicillium species (31.6% of total fungal isolates).

13. The Aspergillus was only represented by A. fumigatus.

Viable Bacterial Community Composition as seen in Molecular Assays

14. The 16S rRNA next-generation sequencing showed the resolution of the bacterial diversity on many phylogenetic levels.

15. The three dominating phyla (~80% of total reads) in both samplings were: Proteobacteria, Firmicutes and Actinobacteria.

16. The distribution of the reads on the class level (~80% of the reads) showed that most of them belong to the class of Bacilli, Gamma-Proteobacteria, or Alpha-Proteobacteria.

17. Molecular analysis (PMA-treated): ~43 genera were dominant (>1%) among total viable bacteria (~120 million reads).

Antibiotic Resistance in Cultivable Bacteria (BSL-2 strains).

18. Bacterial strains isolated from ISS surfaces were tested for their resistance to nine antibiotics using conventional disc method and Vitek 2 system. Most of the Staphylococcus aureus strains were resistant to penicillin. Five strains were specifically resistant to erythromycin and the ermA gene was also detected. The nine-erythromycin sensitive S. aureus strains exhibited spontaneous mutation when rifampin was tested. Some of the S. aureus strains tolerated gentamycin and tobramycin but cefazolin, cefoxitin, ciprofloxacin, and oxacillin inhibited the growth of the S. aureus. Whole genome sequencing (WGS) of 21 ISS strains, exhibiting resistance to various antibiotics, was carried out. The antibiotic resistant genes deduced from the WGS were compared with the resistomes generated directly from the gene pool of the environmental samples.

Antibiotic Resistance in ISS samples using Molecular Methods.

19. Using a targeted amplification panel consisting of over 500 antimicrobial resistance genes, we were able to confirm the results of the phenotypic assays. Specifically, the presence of multiple ß-lactamase genes was observed. The class A gene, tem-1, was present in multiple ISS locations from both flights. A second class A gene, ctx-M-14, was also found in multiple sites of ISS. In addition, presence of mec-A gene was confirmed in several sampling locations from both ISS flights. Finally, the existence of the ermA gene was established. These results suggest widespread and consistent distribution of multiple antibiotic resistance genes throughout the ISS. The resistome data generated via molecular methods will be extremely crucial in determining the microbial significance to the crew health and the ISS maintenance. Whole Genome Sequencing of 21 BSL-2 strains.

20. In an on-going Microbial Observatory experiment on the International Space Station (ISS), multiple BSL-2 bacterial isolates were isolated, identified, and whole genome sequenced (WGS). The genomic data enables the determination of the microgravity influence on pathogenicity and virulence in these microorganisms by comparison to type strains of the corresponding species.

21. Staphylococcus species (16 strains). Staphylococcus isolates were the most prevalent. Staphylococcus aureus was the most abundant in all ISS locations. Although this species is a common human commensal, it causes various types of minor skin infections, bacteremia, or scalded skin syndrome, especially in immunocompromised individuals. In this study, some of the isolates were found to be resistant to erythromycin (IF4SW-P1, IF7SW-P3) and most of the isolates were resistant to penicillin. A few isolates acquired rifampicin resistance during the study (RA isolates).

22. Staphylococcus haemolyticus and Staphylococcus hominis belong to coagulase-negative staphylococci. S. hominis IIF4SC-B9 was resistant to penicillin and erythromycin, but S. haemolyticus IIF2SW-P5 was susceptible to these antibiotics. All three species are reported to be methicillin-resistant by acquiring the Staphylococcal Cassette Chromosome mec (SCCmec), but the methicillin-resistant phenotype was not observed.

23. Pantoea conspicua (1 strain). Pantoea conspicua was originally isolated from human blood. This was the second most prevalent species, and only found in one location during two different flight samplings. P. conspicua isolates were resistant to erythromycin, oxacillin, penicillin, and rifampin.

24. Acinetobacter pittii (1 strain). Acinetobacter pittii is a non-motile coccobacilli isolated for the first time from cerebrospinal fluid. Multiple strains of A. pitti were isolated from the cupola area. The IIF1SW-P1 was resistant to cefazolin, cefoxitin, oxacillin, penicillin, rifampicin.

25. Klebsiella quasipneumoniae (1 strain). Klebsiella quasipneumoniae is a blood-borne opportunistic pathogen. The ISS isolates were retrieved from three locations (cupola, water hygienic convenience compartment [WHC], and exercising machine). The IF3SW-P1 showed resistance to all tested antibiotics except tobramycin.

26. Enterobacter sp. (2 strains). Two multidrug-resistant Enterobacter sp. isolates were found in WHC location. Species of Enterobacter cloacae complex (Ecc) are commonly found in environment, but are of high clinical significance. These two strains are potentially novel and their phylogenetic position is being characterized.

27. Genome sequence analysis. In this study, the draft genomes sequences of 21 strains from the ISS were obtained. WGS sequencing was performed on an Illumina NextSeq instrument with a paired-end module. The A5 assembly pipeline version 20150522 was used to generate draft assemblies applying the default parameter settings. Assembly

	statistics (number of contigs, total genome size, N50 size, median coverage, G+C percentage, and error corrected reads used for assembly) were calculated. The raw reads were in the range of 27 to 82 Mbp per genome. The G+C content was in the range of 31.5 to 38.7% for Staphylococcus species and A. pittii; for other strains the G+C contents were 52.1 to 55.8%. Publications in process: 1. Checinska, A., J. E. Allen, J. Thissen, C. Jaing, K. Venkateswaran. 2016. Draft Genome Sequences of BSL-2 Opportunistic Pathogens, Isolated from the Environmental Surfaces of the International Space Station. Submitted to Genome Announcements. 2. Checinska, A., Q. Tran, V. Stepanov, T. Meyer, F. Karouia, G.E. Fox, K. Venkateswaran. 2016. Insight into the molecular Microbial Diversity of the International Space Station Environmental Surfaces. In preparation. 3. Checinska, A., K.G. Frey, C.L. Redden, J. Thissen, J.E. Allen, C. Jaing, K. Wheeler, K. Venkateswaran. 2016. Targeted amplification of antibiotic resistant genes associated with the International Space Station environment. In preparation.
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Abstracts for Journals and Proceedings	Checinska A. "International Space Station Environmental Microbiome." Postdoc Seminar Series, NASA Jet Propulsion Laboratory, Pasadena, CA, May 7, 2015. Postdoc Seminar Series, NASA Jet Propulsion Laboratory, Pasadena, CA, May 7, 2015. , May-2015
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